

AXIOMOS

The runtime that lets robots change safely while they run.

Admission, rollback, and audit for robot fleets.

Updating a robot fleet is a fire drill.

No rollback. No audit trail. Weeks of revalidation — every release.

THE RELEASE LOOP — EVERY BEHAVIOR CHANGE



Robots receive software **releases**. They do not safely acquire **capabilities**.
One bad update can idle an entire fleet — and most fleets have no rollback path when it does.

1,200+

known critical / high vulnerabilities in the software running most commercial robots (DARPA-funded ROS study) — and nothing gates what it's allowed to do.

Trust-then-run

Today's model: download, trust, execute. No admission check, no rollback plan, no record of who changed what.

The fire drill is about to get much worse.

Models are starting to write the behaviors. Validation can't keep up.

01 - BEHAVIORS FROM MODELS

VLA models generate robot behavior from language — continuously, not quarterly.

RT-2 → π0.5 → GROOT M1 → Helix, 2023-2025. Every brain company is racing to ship behavior faster than any release process can validate it.

4.66M industrial robots live, +9% / yr · IFR

02 - RULES GOT STRICT

New 2025 regulation makes update control and rollback a legal requirement in the EU.

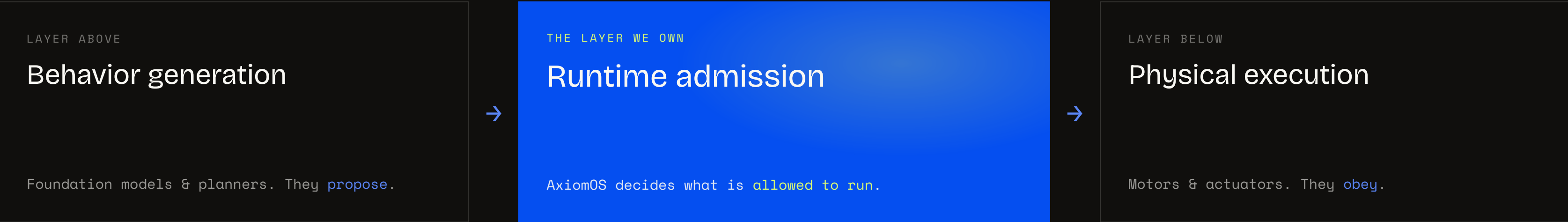
ISO 10218-1/2:2025 · EU Cyber Resilience Act. RaaS contracts make safe OTA and rollback contractual, not optional.

2025 compliance clock already running

*We sell the fix for today's fire drill. The same admission architecture becomes mandatory when models write the behaviors — **and that's already beginning.***

Everyone owns the intelligence. Everyone owns the hardware. Nobody owns the layer in between.

Not a category to believe in — a requirement everyone has that nobody sells.



Nothing touches a motor unless the runtime has decided it's allowed to.

Check what's provable before execution. Enforce the rest at runtime.

DECIDED AT ADMISSION – PROVABLE

Provenance & signature

Resource bounds

Policy compliance

Authority

Static checks with definite answers – decided before a behavior is admitted.

ENFORCED AT RUNTIME – MONITORED

Timing budgets

Speed & joint envelopes

Anomaly detection

Roll back

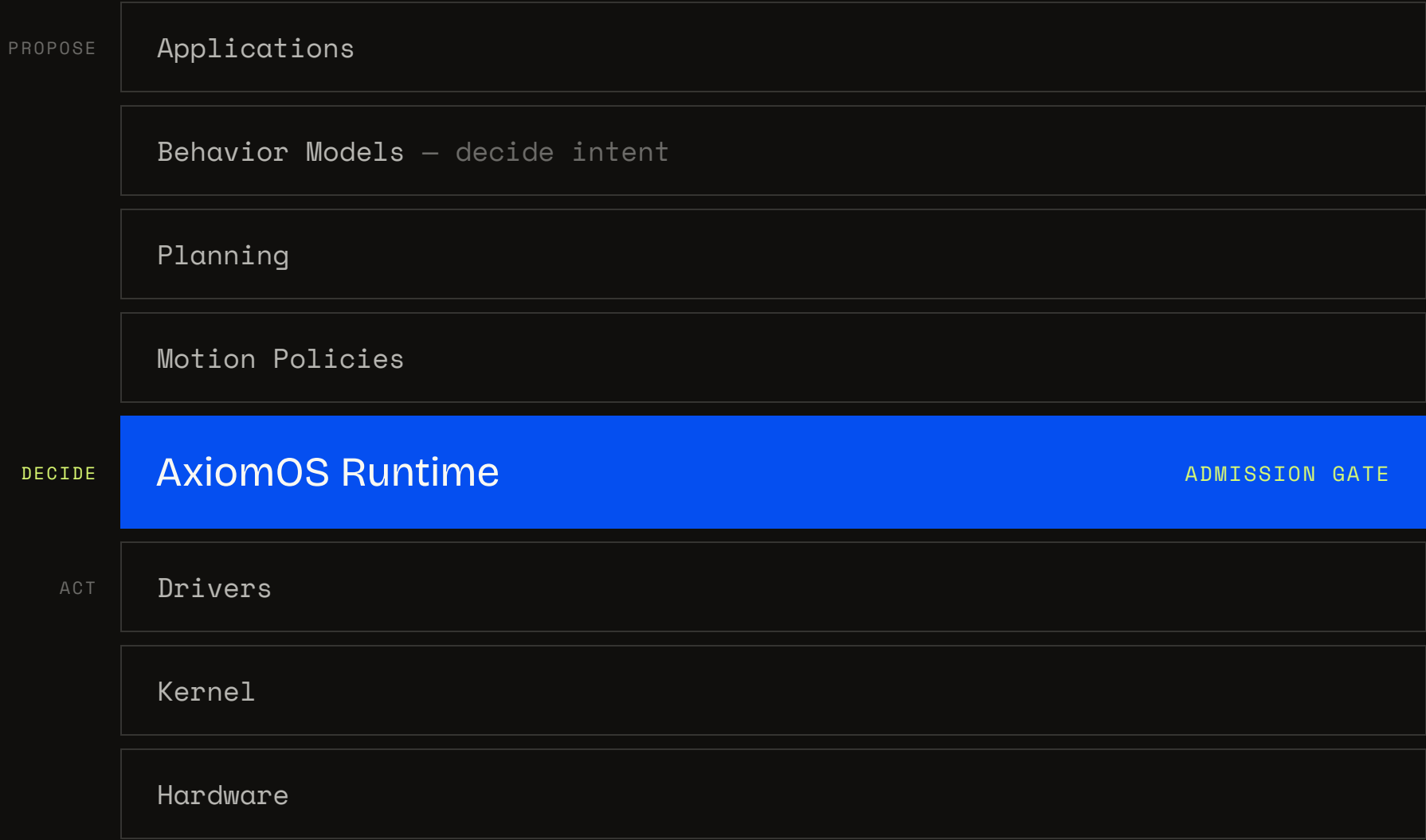
Physical safety depends on the world – so it's enforced live, not assumed at admission.

Beneath both: an independent hardware controller in the actuation path that can always say no.

No gate, no actuation.

Where we sit in the stack.

Between what the robot wants to do and what it's allowed to do.



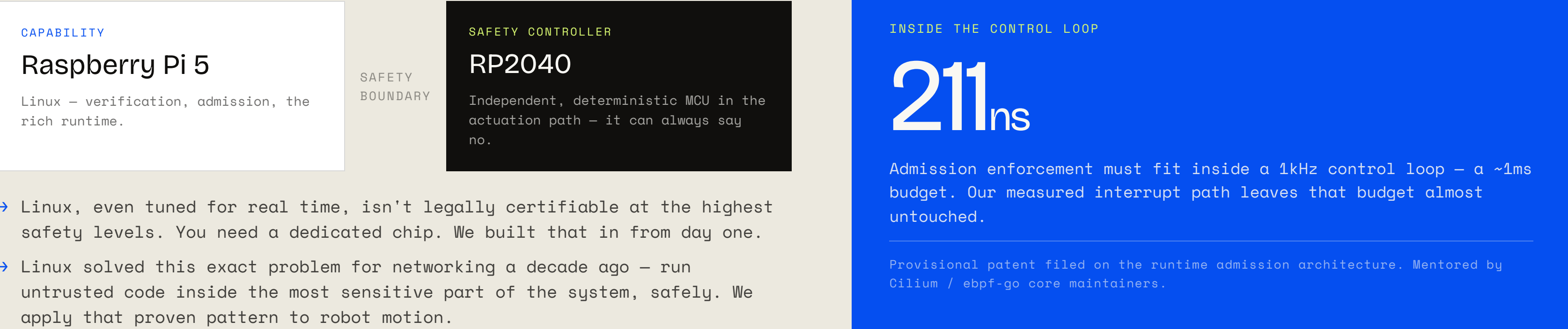
Behavior models decide `intent`. AxiomOS decides `admissibility`.

- Hardware-enforced – an independent controller sits in the actuation path
- Vendor-neutral across every brain and body
- Hardware-spanning – one gate per fleet, not per vendor

Problem validated independently: 2025-26 research (arXiv 2602.04056; “runtime governance layer,” Nov 2025) identifies this same gap. Nobody has productized it.

Two chips, one principle: the safe path is the only path.

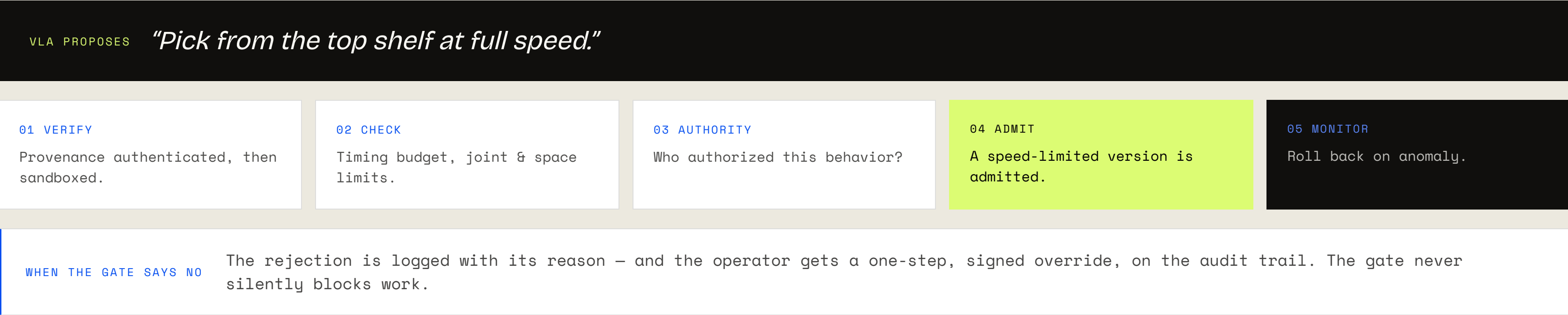
Running end-to-end today on dev hardware — Raspberry Pi 5 + RP2040.



- Linux, even tuned for real time, isn't legally certifiable at the highest safety levels. You need a dedicated chip. We built that in from day one.
- Linux solved this exact problem for networking a decade ago – run untrusted code inside the most sensitive part of the system, safely. We apply that proven pattern to robot motion.

A day in the life of a new capability.

From model output to safe execution – or safe rejection.



The operator sees an `audit trail`: what changed, who authorized it, what was rejected and why. The robot evolves; the operator keeps control.

No fire drill. No black box.

What's proven today — and what this round proves next.

PROVEN

- ✓ Runtime runs end-to-end on Pi 5 + RP2040 – admission, constrained execution, rejection, rollback.
- ✓ Production deterministic firmware in the field securing real assets (Cypherock, 10k+ users).
- ✓ Provisional patent filed on the admission architecture.
- ✓ Mentored by Cilium / ebpf-go core maintainers – references who take calls.

DESIGN-PARTNER PIPELINE

- [N] ROS 2 fleet operators interviewed on their update & revalidation process.
- “[Quote from a fleet operator on what the revalidation cycle costs them – days and dollars.]”
- [Name / class of operator] agreed to pilot on their fleet.

The round's first milestone: convert this pipeline into 3-5 paid design partners.

A small, urgent wedge inside a very large market.

We monetize the runtime, not the robot.

TAM

\$25–30B

Robotics software spend by 2030. ABI – robot software ~\$24.5B.

SAM

\$4–5B

Runtime / middleware + OTA + safety governance today → \$12–15B early 2030s.

SOM · 3 YRS

\$0.5–3M

ARR – 5–50 beachhead fleets. Honest numbers, on purpose.

Morgan Stanley \$5T humanoid by 2050 & Goldman \$38B by 2035 shown only as long-run pull – not as our SAM. Middleware ~\$8.4B (2025); robot OTA ~\$2.8B (2025) → \$9.7B (2034).

Everyone is building the brain or the body. Nobody owns the boundary.

Nobody sells it yet — and we assume everyone adjacent will try.

BRAINS

Physical Intelligence · Figure · Skild · NVIDIA
GROOT — will govern themselves; fleets need an
independent gate

BODIES

Tesla Optimus · 1X · Agility · ANYbotics ·
Dexterity

OPS / OBSERVABILITY

Viam · Formant · Foxglove · InOrbit —
dashboards, not governance

REAL-TIME / SAFETY MIDDLEWARE

Apex.AI · eProsima · RTI — Apex certifies code
before it ships; nobody governs what runs after

THE CONTESTED BOUNDARY

AxiomOS

Runtime admission + governance + rollback. First,
vendor-neutral, and on the certification path —
provisional patent filed.

Apex.AI \$56.5M B · Viam \$30M C · Foxglove \$40M B.

Today the moat is the founder. The rest is a race we intend to win.

The bear cases, named — and how we beat each one.

“NVIDIA BUNDLES IT”

NVIDIA governs NVIDIA. Fleets mixing brains and bodies need a neutral gate that spans vendors — that’s structurally not them.

“THE BRAIN COMPANIES BUILD IT IN-HOUSE”

Self-certified governance is the fox guarding the henhouse. Operators, insurers, and regulators will want independence.

“ROS 2 ABSORBS THE WEDGE”

Open source can ship OTA plumbing; it can’t ship certified, hardware-backed enforcement. Our open core seeds adoption — the gate stays paid.

“APEX.AI EXTENDS DOWN”

Apex certifies code before it ships — build-time. Runtime admission is a different product with a different buyer, and we get there first.

Then it compounds: **certification mappings** encoded at the gate, **deployment evidence** from every admission decision, and a **policy ecosystem** built against our APIs.

Every robot we admit makes the next decision better.

Per-robot licensing that steps up with certification.

Land at tooling prices. Earn the safety premium.

LAND - NOW

Fleet-tooling prices

Design partners pay for the fire-drill fix: safe OTA, rollback, audit – priced against the \$50–200 / robot tooling they already buy.

STEP UP - WITH CERTIFICATION

\$1–3K / robot / yr

The safety-runtime premium follows certification – that's exactly what this raise funds. On-prem runtime + cloud verification.

PLATFORM - LATER

Capability marketplace

Certified third-party capabilities, rev-share – once we own the admission point.

RaaS contracts already make OTA + rollback contractual. Revenue grows with fleet size and capability velocity.

Start where the pain is already real.

ROS 2 / Linux fleets that update often and can't afford to break.

BEACHHEAD

Field / inspection robotics

Warehouse / AMR startups

Mid-market ROS 2 fleets

THE WEDGE

“Ship updates to your fleet without a revalidation fire drill — with rollback and an audit trail.”

MOTION

- 1

3-5 paid design partners
- 2

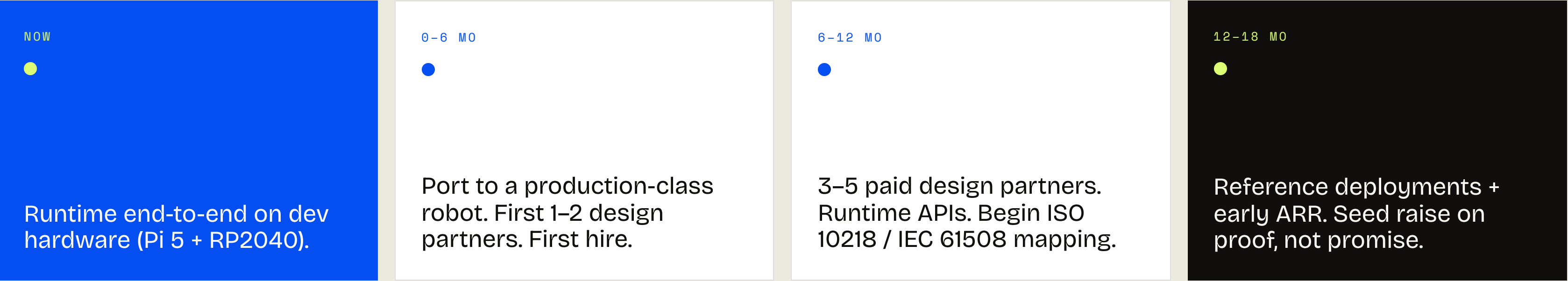
Reference deployments
- 3

Developer adoption — open-core + runtime APIs
- 4

Enterprise

18 months from dev hardware to referenceable fleets.

Milestone-driven, low burn.



Solo technical founder who has shipped the hard part.

Deterministic firmware in production, kernel-level latency, autonomy under failure.

Cypherock

PRODUCTION FIRMWARE

BTC-only firmware — deterministic builds, minimal attack surface.

10k+ users

eBPF on Pi 5

KERNEL LATENCY

Mentored by Cilium / eBPF-go core maintainers.

211ns

ResQTerra

AUTONOMY UNDER FAILURE

GPS-denied SAR drone. DoT 5G hackathon win; 7 founders still build together.

Patent filed

Track record

COMMUNITY + SCALE

Hyperledger LFX mentee · UbuCon India 2025 speaker.

30+ / 10 wins

First two hires are named roles — robotics systems engineer and design-partner engineer — with candidates identified.

\$1.5Mpre-seed

Post-money SAFE · ~18–24 months of runway.

USE OF FUNDS



Milestone: 3–5 paid design partners + reference deployments → seed.

2026 pre-seed norm \$250K–\$2M · SAFE in ~80% of rounds.

The brain is getting cheap. The trust layer is getting essential.

AxiomOS — admission, rollback, and audit for robot fleets.

Runtime running end-to-end on dev hardware today

A requirement every fleet has — that nobody sells yet

Funding 18 months: proof-of-skill → proof-of-business